Predicting CO2 levels using Bluetooth to estimate and predict the occupancy of rooms

What?

1. We will be measuring and predicting occupancy by using Bluetooth Technology to find devices and filter them, s.t. we only consider mobile devices to get accurate representation of participants.
2. Then we will be estimating CO2 levels using mathematical modelling which will enable us to deploy “virtual sensors”. These can be used per room to estimate airflow needed to enable optimal productivity and wellbeing levels. Additionally, individuals can run a standalone personal mode

Inputs?

1. Room size
2. Live Occupancy & Predicted occupancy (take into account special dates like exams, events etc.?
3. Number of Windows => make customizable
4. Window size in m^2 -> still needed
5. Time elapsed since window were closed
6. How long were the windows open the last time? => needs to be added
7. Simulated data on occupancy for the last 7 or 30 days
8. Room Calender Schedule

*note: might need filtering if bluetooth values change rapidly

Outputs (what we make visible)?

1. **Live & Forecasted CO2 Curve (ppm):** Real-time room concentration plus a 4-hour forward projection driven by predicted occupancy.
2. **Cognitive Performance Index (%):** Live translation of ppm into executive function impact (e.g., *"-24% strategic decision-making"*), turning invisible gas into a tangible productivity metric.
3. **Actionable Ventilation & Schedule Alerts:** Exact countdowns for when to open windows and for how many minutes (e.g., *"Open 2 windows at 14:45 for 6 mins"*), plus automated 10-minute room-flush buffers inserted between back-to-back calendar bookings. => add alrts

Assumptions?

1. **Tenant-controlled ventilation:** Our strategy centers on manually opening windows and refining timetables, particularly since over four-fifths of European and UK commercial properties—including rented spaces, older university buildings, and modular rooms—lack **direct control over central HVAC systems** or depend solely

on natural airflow. Additionally, background air leakage with closed windows is treated as a steady baseline rate (0.2–0.4 ACH).

2. **Zero-CapEx Deployment:** Companies leasing office space will not pay to wire, mount, and later de-install hardware CO2 sensors across dozens of rooms, making zero-hardware Virtual Sensors the only scalable B2B model.

Architectural Design

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[STAGE 1: Occupancy Engine]
  Inputs:
  • Live BLE device count (current state)
  • 7–30 days historical occupancy patterns
  • Room Calendar / Timetable (optional context)
        │
        ▼
  Output of Stage 1 / Input to Stage 2:
  ► PREDICTED OCCUPANCY CURVE N(t) (e.g., next 4 hours, in 5-min steps)
        │
        ▼
[STAGE 2: Virtual CO2 & Airflow Engine]
  Inputs:
  • Predicted Occupancy N(t)  <-- Drives future CO2 generation G(t)
  • Room size (Volume V) & Number of windows
  • Last window open duration & timestamp (Sets initial C_0)
  • Indoor Temp + Outdoor Weather (Sets airflow Q)
        │
        ▼
  Final Actionable Outputs:
  • Live CO2 & Cognitive Throughput Index
  • "Next Window Open" Countdown (e.g., "Open at 14:40 for 5 mins")
  • Optimal Ventilation Schedule for the rest of the day
```

skip temperature

For the presentation:

Key Quantified Research Findings + what comes up on Google when searching for the study (good opener?)

- **Harvard T.H. Chan School of Public Health (COGfx Study):** In double-blind crossover trials testing knowledge workers across varying office air conditions, cognitive function scores were **61% higher** at 950 ppm CO2CO2 and **101% higher** at 550 ppm CO2CO2 compared to conventional office conditions (around 1,400 ppm). The functional domains showing the sharpest drops at higher CO2CO2 were high-order executive functions:
    - **Crisis Response:** ~97% decline at 1,400 ppm vs. 550 ppm.
    - **Strategy:** ~91% decline.
    - **Information Usage:** ~85% decline.
    - *Note:* Simple task orientation and raw information search suffered much less, indicating that routine tasks survive high CO2CO2, but complex problem-solving, architectural design, and deep work collapse.
- **Response Time and Processing Throughput (Harvard Global Study):** A follow-up study evaluating real-time cognitive responses found that for every **500 ppm increase in CO2CO2**, participants experienced:
    - **1.4% to 1.8% slower response times** on cognitive assessments.
    - **2.1% to 2.4% lower throughput** (correct responses per minute).
- **Lawrence Berkeley National Laboratory (Satish et al.):** Evaluating decision-making performance via the Strategic Management Simulation (SMS) tool, researchers observed moderate drops in performance starting as low as **1,000 ppm**, and dramatic, statistically significant reductions across nearly all decision-making metrics when levels reached **2,500 ppm**.

- "We don't claim to measure an individual's personal IQ in real time—we calculate the **Environmental Cognitive Ceiling**based on the Harvard T.H. Chan School of Public Health COGfx empirical curve. Moreover, because our virtual sensor models real human occupancy in unventilated rooms rather than synthetic gas, our CO2 curve tracks the full buildup of human bioeffluents and thermal load. Just like CPU thermal throttling, it tells a team when their room environment is throttling their strategic decision-making capacity by 25%."

Occupation prediction

- trained on historical occupancy data looking for patterns, eg fridays are always going to be the least busy working day
- short term predictions use gradient boosting, which combines lots of small decision trees to learn relationships between the current occupancy, recent changes and time of day/week

- [We also compare this against linear (ridge) regression and seasonal statistical models to check whether this approach actually improves accuracy]
- Provides a 24hr forecast with approximate uncertainty ranges

More data for the presentation:

https://docs.google.com/document/d/1uaQ7Ty1s7-w1sVKUIL67Ltrpe_lLzftJRDmKop885Ml/edit?usp=sharing

Physics-based mass-balance model,

Name ideas:

- Regio => Latin for environment
- Vicinia => similar
- Ambitus => similar
- Aptare => adjust in Latin
- Calum => Air in Latin

To-Do:

- Check Formula correctness for Domainc Specific breakdown and Overall Congnitive Performance Score => should be ok
- add admin panel to change distance of arduino Done
- Optional, increase uncertainty further along the 24hr forecast
- Occupancy Forecast for Demo
- Optional:

1. **Call BLE Sniffing "Alternative Data for Nowcasting":**
    - *The Pitch:* "In quant finance, you use satellite imagery of parking lots to predict retail earnings before the quarterly report. We do the exact same thing for physical spaces: Calendar bookings are 'reported intent' (which is often wrong due to ghost meetings or overcrowding), while our BLE packet sniffer acts as **real-time alternative data** to correct the baseline and predict true occupancy."
2. **Add Confidence Intervals (Uncertainty Bands) to Your Forecast Curve:**
    - Quants hate single-line point predictions because real data is probabilistic.
    - *Quick UI Upgrade:* On your 4-hour forecasted CO_2 chart, plot your main predicted curve, plus a shaded $\pm 1\sigma$ **confidence band** around it (e.g., upper bound if +2 extra people walk in; lower bound if someone leaves early). That single visual touch screams "quantitative rigor."
3. **Frame the Output as "Expected Value (EV) / Risk Optimization":**
    - *The Pitch*: "Opening a window in winter has a thermal cost (heating loss), while keeping it closed has a cognitive cost (executive function drop). Our engine solves an **optimization problem**: finding the exact timestamp t and duration Δt (e.g., *Open at 14:45 for 6 mins*) that maximizes cognitive throughput while minimizing thermal disruption".

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Other

Useful Outputs

- knowing cold spots in a building to determine where should be upgraded
- Digital twin of office environment
    - CO2 output
    - Foot traffic
    - When to increase ventilation
    - Temperature

Giving advice on where to go to evenly distribute people within a space